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MTH101 - Calculus And Analytical Geometry

x^2-9|=.....
- |(x-3)^2|
- |(x+3)^2|
- ***|x-3||x+3|***
- |x+3||x+3|

Usually the number that signifies the idea of f(x) being as close to limit L as want to be must be a/an .....
- ***Integer***
- Natural number
- Small positive number.
- Small negative number.

A function f is said to be continuous on a closed interval [a, b] if f is continuous from the right at "a" and "f" is continuous from the left at "b" and "f" is continuous on
- (a,b]
- [a,b)
- [a,b]
- ***(a, b)***

If f is continuous on [a, b], and if f(angel) and f(beer) have opposite signs, then there is ..... one solution of the equation f(x) = 0 in the interval (a, b).
- at most
- exactly
- ***at least***
- not more than

(epsilon) used in the definition of limit can be a negative number.
- True
- ***False***

If a function is differentiable at a point then it is continuous at that point. The converse is
- False
- ***True***

If the function f and g are continuous at c, then f + g is ..... at c.
- Discontinuous
- ***Continuous***

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If f is continuous on a closed interval [a, b] and C is any number between f(angel) and f(beer), inclusive, then there is at least one number x in the interval [a, b] such that ---------

- f(x) is not equal to C
- ***f(x) = C***
- f(x)>C
- f(x)<C

|x-3| < 1 implies.....

- -4 < x < 4
- ***2 < x < 4***
- -2 < x < -4
- x-3 < 1

If for any positive number e(epsilon) we can find d (delta) such that| (3x-5) - 1| < e ,if x satisfies 0< |x-2| < d Then f(x) =.....

- 3x-5- 1
- x-2
- ***3x-5***
- None of these

A graph of an equation is the points on the xy- plane that .....the equation.

- ***Satisfy***
- Does not satisfy.

If two lines have same slope, say 1, then these two lines are _____

- ***Parallel to each other***
- Parallel to x-axis
- Parallel to y-axis
- Perpendicular to each other

There is one-to-one correspondence between the points on co-ordinate line and _____

- Set of natural numbers
- Set of integers
- Set of irrational numbers
- ***Set of real numbers***

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Which of the following is y-intercept of a line represented by equation, 5y = - 2x + 16

- 2
- 5
- 8
- 16 / 5

Does the point (1,2) satisfies the equation: 2y=x2−1

- Yes
- No

Let x is a real variable. The inequality -5 < x < 2 denote_____

- Set of all rational numbers between -5 and 2
- Set of all real numbers between -5 and 2.
- Set of all natural numbers between -5 and 2
- Set of all integers between -5 and 2

Which of the following option describes the graph of equation:(x2)+(y2)=1

- A parabola whose vertex is at point (1, 0).
- A straight line whose slope and y-intercept is 1.
- A circle centered at origin and its radius is 1
- None of these

Graph of the equation x2+y2=9 represents a.....

- ***Circle not sure***
- Parabola
- Ellipse
- None of these.

Which of the following is distance between the points (2, 5) and (-1, 1)?

- Square root of (-7)
- Square root of (37)
- 25
- 5

The inequality, {1 / x} < 1 can be simplified to which of the following?

- (1 - x) / x > 0
- (1 - x) / x < 0
- (1 – x) < 0
- (1 – x) > 0

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If Sin(3x^2) / 6 + C is the anti–derivative of a function f(x), then f(x) = _____.

- x^2 Cos(3x^2).
- ***x Cos(3x^2).***
- x Cos(3x).
- none

Which of the following is the integral of Sin(3x+5) with respect to x ?

- -1/3[-Cos(3x+5)]
- ***1/3[-Cos(3x+5)]***
- 1/15[-Cos(3x+5)]
- -Cos(3x+5)

If 'n' goes from 1 to 3 and the summation of 'na' = 6a, then the value of 'a' is -------

- ***6***
- -6
- 1
- undetermined

If 'n' goes from 1 to any large ODD number then the summation of '(-1)^n' = -------

- ***-1***
- 0
- 1
- that specific large ODD number

1+2+3.....+t equals

- n(n+1)/2
- ***t(t+1)/2***
- n(n+1)(2n+1)/6
- none

Question No: 25 (Marks: 3)

Discuss the concavity of the function $f(t)=-2(t-1)(t-2)+3t^2$ on any interval using second derivative test?

Question No: 26 (Marks: 5)

Differentiate w.r.t. x by chain rule $y=\cos^2(x^3)$

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Question No: 27 (Marks: 10)

Compute derivative of trigonometric function $Sin(x)$ by definition.

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